

Exam Date & Time: 02-Dec-2024 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VII Semester - End Semester Examination - Nov / Dec -2024

FINANCIAL MATHEMATICS [NPT-3207]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) The discount rate for INR 7000 loan for 60 days (take 360 days in a year) if the borrower gets INR 6790 is? (5)
 - A)
 - B) Ramesh would like to borrow INR 2500 from his local bank for 2 years at a discount rate of 9%. The proceeds Ramesh would receive and the discount amount received by the bank would be respectively? (5)
- 2) An enthusiastic new engineering graduate plans to start a consulting firm by borrowing Rs. 100000 at 10% per year interest. The loan payment each year to pay off the loan in 7 years is approximately (5)
 - A)
 - B) A father, on the day his son is born, wishes to deposit an amount that would provide withdrawals of Rs. 20000 on each of the son's 18th, 19th, 20th, and 21st birthdays. At interest of 12% per year, the amount to be deposited will be? (5)
- 3) Effective annual interest rate for nominal interest rate 14% compounded semiannually is? (5)
 - A)
 - B) Single amount at the end of fifth year, which will be equivalent to uniform annual series of Rs. 20000 per year for 12 years at 7% interest rate compounded annually will be Rupees? (5)
- 4) The owner of a local pizzeria needs Rs. 250000 in 5 years to open another branch. Total deposited in the company's sinking fund that earns 8% compounded monthly is? (5)
 - A)
 - B) An asset had first cost of Rs. 60000 and salvage value of Rs. 5000 after 11 years. If it is depreciated by the basic declining-balance method using a rate of 10%, the depreciation charge in the tenth year will be? (5)
- 5) A publishing company estimates that 13000 books will be sold per year with total fixed cost of production as Rs. 701250. The average cost of materials and the direct labour cost are Rs. 325 and Rs. 102.50 per book, respectively. The company sells the book for (5)

Rs. 510.

- A) The break-even level (in terms of number of books produced) will be?
- B) A man invested in 500 shares of a local business stock selling for Rs. 15.75 a share and pays Rs. 70 plus 0.006 on this investment. Later, he sold half of his shares at Rs. 17.25. (5)
The investment rate of return and capital gain is?

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